

ECEN 403(H) Individual Written Technical Report

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Abstract

Our capstone project investigates how lunar imagery can support terrain reconstruction and rover navigation evaluation. My work has focused on preparing image information and developing the reconstruction portion of the workflow. I first generated SAM2 masks for the team's initial pipeline, which is now planned as a backup pipeline. At my advisor's request, I then used Astra as a development tool to construct a more robust reconstruction pipeline using the available Chandrayaan-3 imagery and calibration records. The resulting implementation performs stereo reconstruction, conservative filtering, local meshing, and registration where observations provide sufficient overlap. The final filtered batch produced 50 local point clouds and 50 partial meshes containing approximately 4.97 million per-frame points, while registration accepted 17 of 129 candidate links. Numerical and reproducibility tests support implementation correctness, but external validation of physical terrain accuracy has not been fully completed. The reconstructed surfaces also remain incomplete because the source imagery does not provide continuous coverage. This report describes my contribution, current progress, remaining limitations, and the next steps needed to connect reconstruction to the team's navigation work.

Project Context and Personal Contribution

The team's overall objective is to develop a real-to-simulation workflow that connects lunar imagery to terrain reconstruction and eventually to rover navigation experiments. Each stage depends on the previous stage producing acceptable outputs. A navigation experiment requires more than a visually convincing terrain model; the simulated terrain must meaningfully represent the geometry of the actual lunar environment. My reconstruction work addresses this requirement by converting available lunar imagery into geometry that can eventually be evaluated and incorporated into simulation.

My contributions so far have developed in two stages, one for our initial reconstruction pipeline concept, and the other for our current concept. Our initial pipeline concept used statistics describing lunar terrain, such as the frequency, size, and spatial relationships of rocks and craters, together with low-fidelity orbital elevation data to create generated terrain that is informed by, but not identical to, real lunar terrain. After some discussion, we pivoted to attempting to directly generate meshes from Chandrayaan-3 stereo images only, and the statistically informed pipeline became our backup.

For the first pipeline, I initially developed a system to generate SAM2 [1] masks of rocks and craters in the Chandrayaan-3 images; this allowed us to know where rocks and craters were located within the images, and, combined with depth data from stereo reconstruction, we could determine the size of the objects and their spatial relationships to one another. After reevaluating our pipeline and pivoting to direct reconstruction, I was instructed to use Astra as a tool to aid in reconstructing a robust pipeline. Once this pipeline was completed, its geometric assumptions and intermediate results were examined.

Initial Work: Mask Generation Using SAM2

My first technical contribution was SAM2 mask generation for the first pipeline design, which is detailed in **Figure 1**. As previously mentioned, the first pipeline focused on plausible terrain generation and not direct reconstruction; as such, statistics on the frequencies of different sized features would need to be collected. The original feature-processing plan included SAM2 labeling, human classification, conversion to YOLO labels, and YOLOv8 training. Following advisor feedback to keep feature detection simple, I pivoted solely to generating masks with SAM2 without later YOLOv8 training. Initial exploration and tuning yielded results such as what is shown in **Figure 2**.

In the current state, crater-like regions were segmented out and masked due to shadow contrast, but rocks, especially small ones, posed a challenge because they blended into the surrounding terrain. Since the team decided to pivot to direct terrain reconstruction, work on mask generation and feature statistics has been put on hiatus; however, it remains in a stable state as a backup in the case that the team needs to pivot back to this pipeline. In that case, mask generation would need to be tuned to more accurately mask out craters and rocks, and then the verified feature masks would be associated with stereo geometry. Once sizes and spatial relationships were established, statistics would be generated that could feed a terrain generation model.

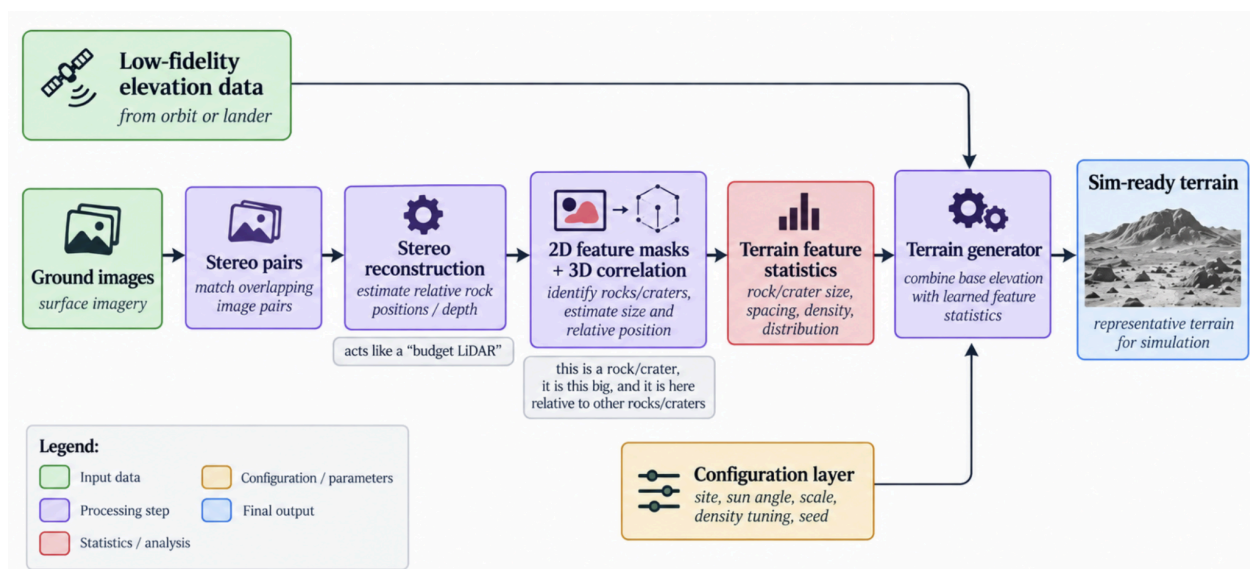


Figure 1. Initial terrain generation pipeline proposal.

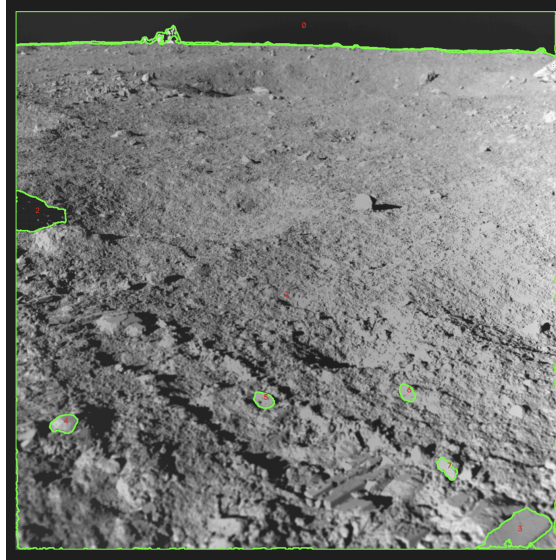


Figure 2. Example SAM2 segmentation on Chandrayaan-3 surface imagery.

Dataset Investigation and Assumptions

Once the team decided to pivot to direct terrain reconstruction from Chandrayaan-3 [2] stereo images only, a more thorough investigation of the dataset was conducted to determine feasibility of reconstruction and the ideal approach. The image dataset contained 56 candidate observations, each represented by raw and radiometrically calibrated left and right images, for 224 images total. Each image was 1024 by 1024 pixels and contained 8-bit monochrome data. Observations were sparse still imagery rather than continuous video; observations spanned August 23 through September 2, 2023, with a median gap of 3 hours 24 minutes between observations. The CAHV camera models supplied in Chandrayaan-3 documentation were also examined. These models encode the camera centers and projection vectors [2].

There were several initial limitations identified. Firstly, individual images lacked XML labels, and available documentation did not provide a complete per-image pose stream. Raw images also sometimes included severe saturation, shadows, and glare, which could limit recoverable information even when processed successfully.

Observation timing was also a significant consideration. Candidate left and right images were typically separated by approximately 30 to 43 seconds. The fixed camera geometry supplied in Chandrayaan-3 documentation can thus only be used under an assumption that the rover was stationary during this time. Good image alignment provides some support for this assumption, but it cannot completely rule out rover motion. For example, translation parallel to the camera baseline could change the depth scale without producing a corresponding vertical epipolar error. Therefore, there is a physical uncertainty that can't be accounted for simply by improving the pipeline's ability to match features; however, in the future, this uncertainty may be able to be resolved through discovery of additional Chandrayaan-3 documentation or contact with the Chandrayaan-3 team.

Reconstruction Pipeline and Experimental Development

The new reconstruction pipeline follows a conventional stereo reconstruction sequence: camera model interpretation, image rectification, disparity estimation, depth calculation, filtering, point cloud export, and local meshing. It then attempts registration between observations with sufficient shared features.

Camera projection is derived from the CAHV parameters found in Chandrayaan-3 documentation. Rectification transforms the two images into a common viewing geometry so that corresponding points lie approximately on the same image row [3]. After rectification, disparity represents the horizontal separation between corresponding points. Depth is calculated using

$$Z = \frac{fB}{d} \quad \text{Equation 1}$$

where Z is depth, f is focal length, B is the stereo baseline, and d is disparity [4].

Dense matching uses OpenCV's StereoSGBM implementation [5]. Experiments were completed to compare raw imagery, radiometrically calibrated imagery, and contrast enhancement. On one observation (Observation 003), the initial raw run retained 123,115 points, while the calibrated run retained 148,250. Contrast enhancement retained 129,904. These comparisons show differences in coverage, but increased point count alone does not establish improved accuracy. For example, radiometric processing can obscure the appearance of clipped raw pixels without restoring information lost during acquisition.

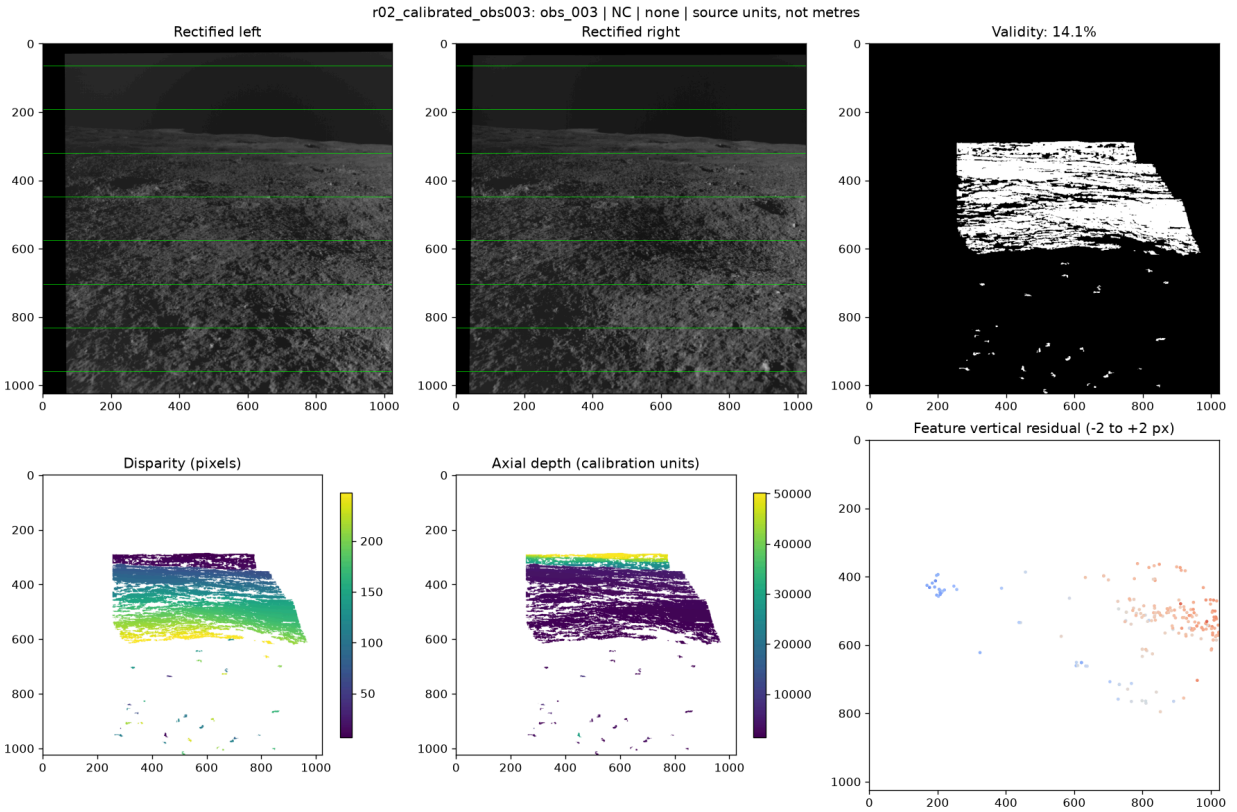


Figure 3. Stereo reconstruction on Observation 003 using radiometrically calibrated imagery.

The original 256-disparity configuration limited reconstruction of nearby terrain; to combat this, a wider 512-disparity configuration was adopted. For example, on one observation (Observation 029), the wider disparity configuration increased retained points from 99,390 to 134,544, a 35.37% increase, while image coverage rose from 9.48% to 12.83%. Of the retained points in the wider run, 14,960 had disparities of at least 256 pixels, demonstrating recovery beyond the original search range. The validity and disparity panels show where this additional support occurred. However, the wider configuration also introduced image padding and adjusted filtering thresholds, so the overall increase cannot be attributed solely to search width. These results demonstrate increased reconstruction coverage, not independently verified improvements in geometric accuracy.

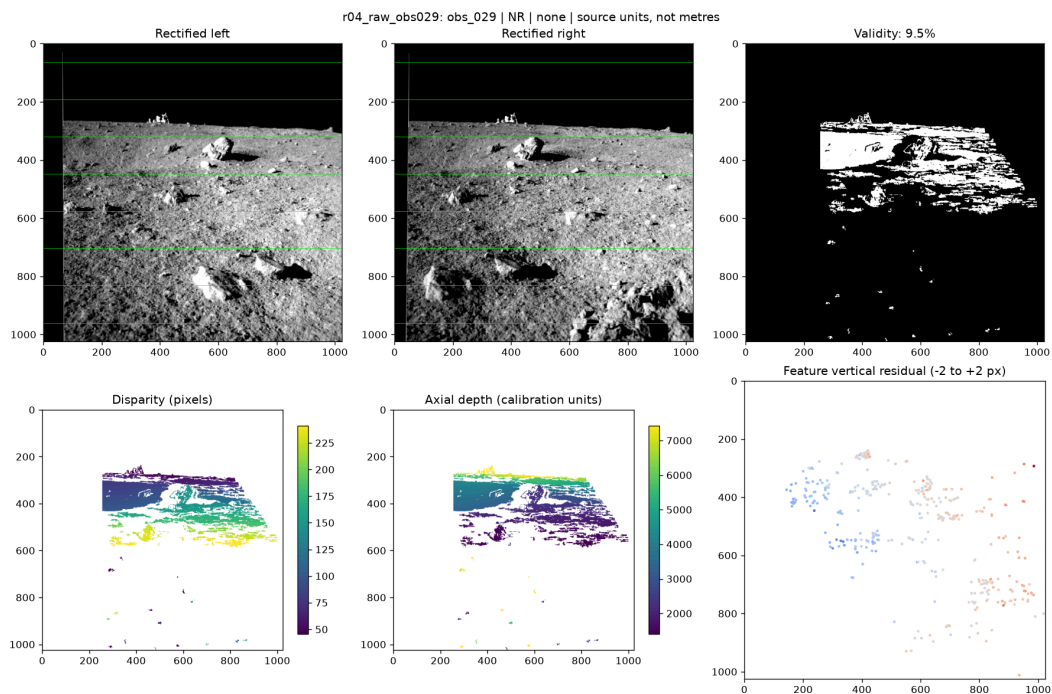


Figure 4. Observation 029 reconstructed using the 256-disparity configuration.

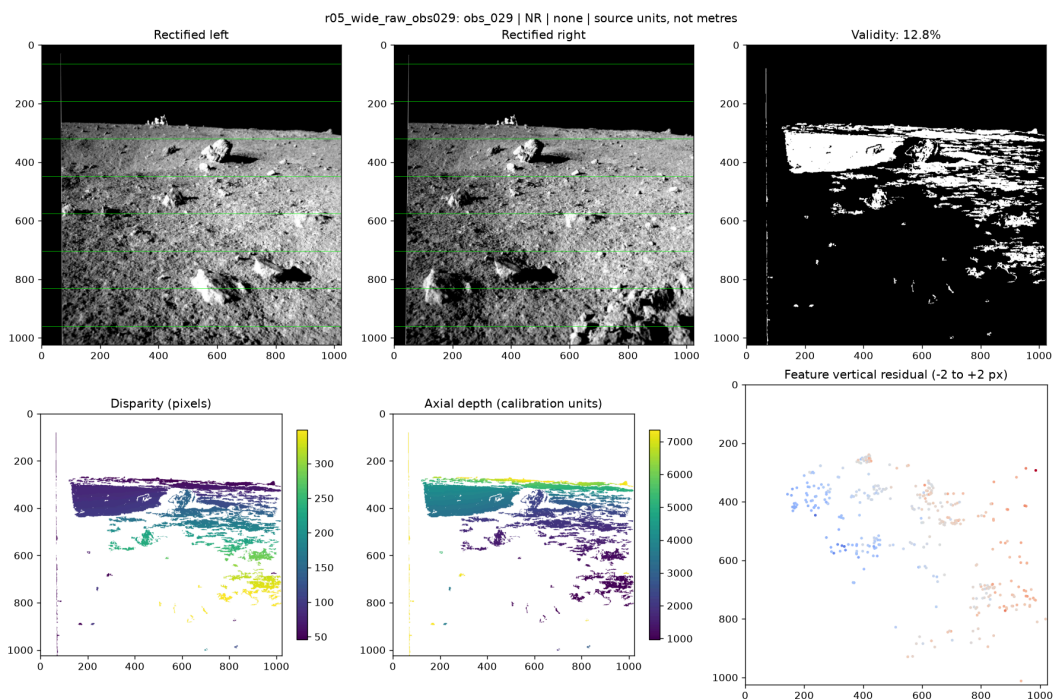


Figure 5. Observation 029 reconstructed using the 512-disparity configuration.

Reliability checks compare left-to-right and right-to-left disparities and reject pixels that disagree by more than one adjacent pixel. Additional masks exclude unreliable raw intensity regions and regions of weak texture. Final filtering removes unsupported borders, isolated regions, and areas with inconsistent disparity. This trades coverage for stronger local support; however, it can also remove genuine small terrain features, so the filtered output is not a complete obstacle inventory.

Local meshing connects nearby supported samples while rejecting triangles that cross excessive depth changes or long edges. It does not fill gaps with generated surfaces. This preserves a useful distinction between terrain supported by observations and terrain that would have to be inferred. The resulting meshes are partial surfaces; open boundaries and holes are expected consequences of the available evidence.

Registration extends reconstruction beyond individual image pairs. Feature correspondences and source stereo depth support pose estimation between candidate observations. A held-out subset of correspondences evaluates reprojection after fitting, while target stereo depth provides another consistency check. Only accepted links contribute to connected components. Disconnected observations remain separate instead of receiving invented positions in a global map.

Current Results and Scope of Validation

The final filtered batch contains 50 local point clouds and 50 partial meshes from 56 candidate observations. It totals 4,969,574 per-frame points and 508,095 local mesh triangles, with median coverage of 10.03%. These totals include overlapping observations. They are not counts of unique points in a single global terrain model. Six candidates were rejected because the available evidence failed the geometric or image-quality criteria.

Registration tested 129 candidate links and accepted 17, producing six multi-frame components and leaving 31 observations isolated. These results establish that some local overlap supports registration while also showing why the current archive does not support a continuous global reconstruction under the implemented criteria. The registered meshes combine existing observed triangles; they retain holes and may contain overlapping faces.

It was found that there was maximum vertical round-trip error of approximately 5.68×10^{-13} pixels and maximum three-dimensional round-trip error of 2.18×10^{-11} mm. A synthetic image pair with a known 24-pixel shift returned the expected disparity signs. These small errors support implementation correctness on the constructed tests.

Export and reproducibility checks address the reliability of the saved artifacts. A repeated baseline run reproduced the original left and right disparity, depth, and XYZ arrays exactly, including invalid value locations. This confirms that results are reproducible.

Together, these results support an operational and auditable reconstruction pipeline. However, they do not establish that the results are simulation-ready. We will need to devise methodology to fill gaps and validate the completed terrain prior to considering the results simulation-ready.

Relationship to Overall Team Progress

The team's navigation work currently supplies a complementary simulation baseline. The navigation team successfully set up ROS2 Jazzy and Gazebo Harmonic; they used them to complete three fixed rover routes with final goal errors of approximately 0.277–0.298 meters and no recorded obstacle contacts. Additionally, fault injection cases resulted in appropriate stop outcomes. The navigation team also reports passing stationary RGB-D interface tests at optical distances of 1, 2, and 4 meters. These results are promising and indicate that we are on track to integrate the reconstructed 3D terrain into the environment when it is ready.

The current navigation tests use a known obstacle map and have access to ground-truth odometry. The next steps on the navigation side will be to insert the reconstructed lunar terrain into the simulation so the project can demonstrate camera-driven navigation across reconstructed terrain. My immediate integration responsibility concerns the terrain side of that connection.

Next Steps and Acceptance Criteria

For our primary pipeline (direct terrain reconstruction), the first priority is external validation. A measured target or trusted terrain reference is needed to evaluate physical error. Registration should also be assessed with attention to duplicate images and whether apparently repeated support provides genuinely independent geometry. Once the primary pipeline is validated, additional steps will most likely be required to fill holes and unknown regions. Additional investigation and discussion is necessary to determine how.

For the simulation, I will work with the navigation and integration teams on one selected local mesh initially. Acceptance will require a documented origin, axis convention, scale, gravity alignment, and sufficient mesh quality. Visual and collision geometry will also need to be reviewed. Successful file import will be a compatibility check, not the final acceptance criterion. Once this mesh is prepared and properly set up within a Gazebo world, the navigation team can begin experiments. While they are working with the first approved mesh, I will prepare additional high-fidelity meshes to be used while factoring in their feedback.

For the backup pipeline, the next step is to fine tune the approach to identify smaller features and then review and classify a representative set of SAM2 mask proposals. Evaluation will record missed objects, incorrect masks, mask quality, and duplicate proposals. After this review, accepted masks will be associated with valid stereo depth and used to generate statistics for the terrain generator. However, development of this alternate pipeline will remain on hiatus unless it is needed.

Conclusion

My work progressed from SAM2 mask generation for a terrain generation pipeline to development of an auditable stereo reconstruction workflow. The present implementation produces supported local geometry, records rejected cases, and passes specific numerical, consistency, and reproducibility checks. Its main remaining limitations concern acquisition motion and incomplete surfaces. Resolving those limitations is the necessary next step toward terrain that can support meaningful rover navigation evaluation.

References

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